

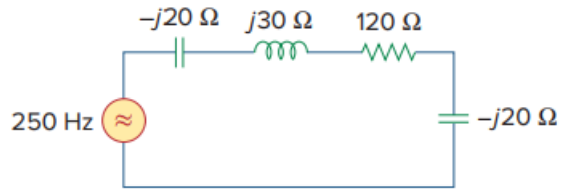
Problem

A series audio circuit is shown in Fig. 9.88.

(a) What is the impedance of the circuit?

(b) If the frequency were halved, what would be the impedance of the circuit?

Figure 9.88:



Step-by-step solution

Step 1 of 2

$$(A) \quad Z = -j20 + j30 + 120 - j20$$

$$Z = 120 - j10 \, \Omega .$$

Step 2 of 2

(B) If the frequency were halved, $\frac{1}{\omega_c} = \frac{1}{2\pi f_c}$ would cause the capacitive

Impedance to double, while $\omega_L = 2\pi f L$ would cause the inductive impedance to halve. Thus,

$$Z = -j40 + j15 + 120 - j40$$

$$Z = 120 - j65 \, \Omega .$$